

chapter7 Wireless and Mobile Networks

 maybe粗整理，大手笔斧正！

 第一部分是Wireless

1. Outline

1.1 BackGround:

- 2019: Mobile subscribers ($10\times$ fixed-line); mobile-broadband devices ($5\times$ fixed-broadband).
- 4G/5G uses internet protocols (e.g., SDN).
- Core challenges:
 - a. Wireless: Wireless link communication.
 - b. Mobility: User network attachment switching.

1.2 Elements of a wireless network 要素

1. **Wireless hosts:** Devices (laptop, smartphone, IoT) running apps; can be stationary/mobile (wireless $\neq$ always mobile).
2. **Base station:** Connected to wired infrastructure; relays packets between wired network and wireless hosts (e.g., cell towers, Wi-Fi access points).
3. **Wireless link:** Connects hosts to base stations (or as **backbone**); uses **multiple access protocols**; varies in rate, distance, frequency.

标准 / 技术	类型	吞吐量	典型范围
Bluetooth	短距无线	2 Mbps	Indoor (10-30m)
802.11b (Wi-Fi 2)	Wi-Fi	11 Mbps	Indoor (10-30m)
802.11g (Wi-Fi 3)	Wi-Fi	54 Mbps	Indoor (10-30m)
802.11n (Wi-Fi 4)	Wi-Fi	600 Mbps	Outdoor (50-200m)
802.11ac (Wi-Fi 5)	Wi-Fi	3.5 Gbps	Outdoor (50-200m)
802.11ax (Wi-Fi 6)	Wi-Fi	14 Gbps	Outdoor (50-200m)
802.11be (Wi-Fi 7)	Wi-Fi	46 Gbps	-
4G LTE	蜂窝网络	600 Mbps	Midrange (200m-4Km)
5G	蜂窝网络	~10 Gbps	Midrange (200m-4Km)
802.11 af/ah	长距 Wi-Fi	600 Mbps	Long range (4Km-15Km)

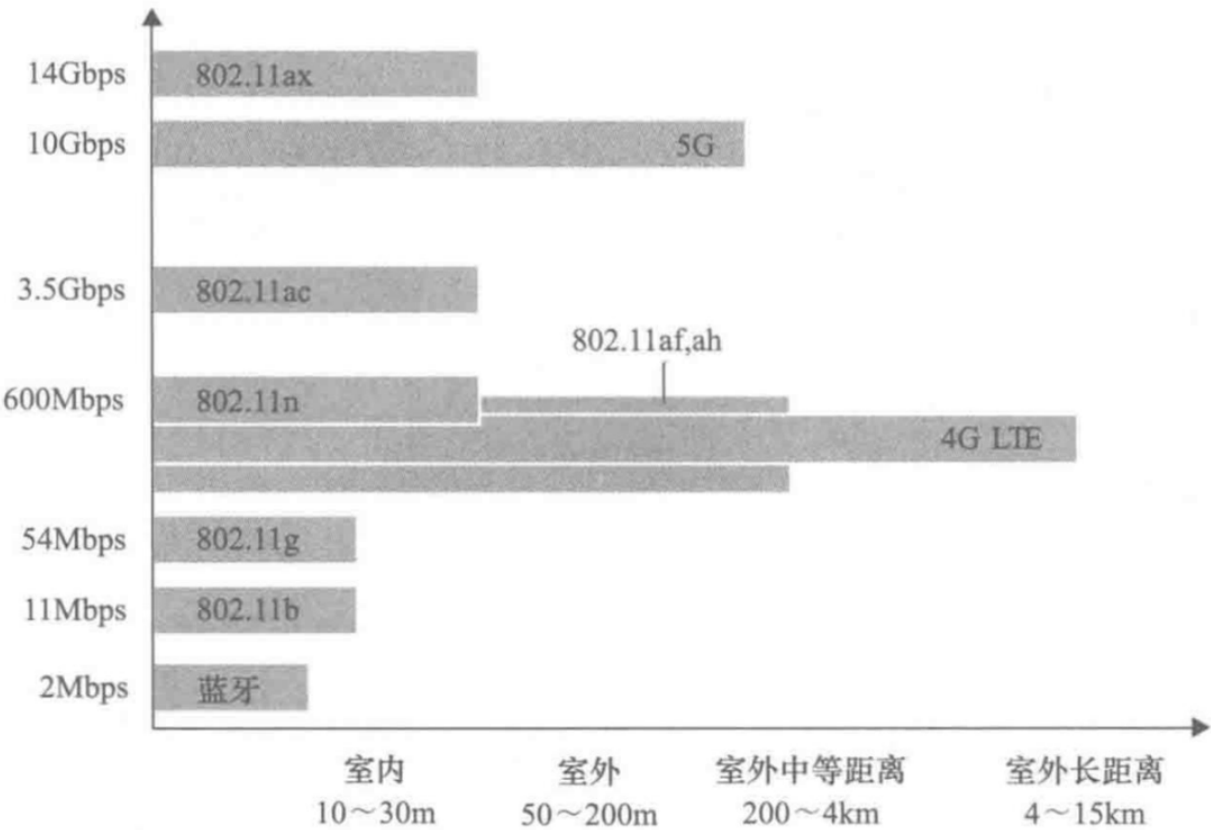


图 7-2 WiFi、蜂窝 4G/5G 和蓝牙标准的无线传输速率和范围（坐标轴不是线性的）

1.3 wireless network modes and classifications

1.3.1 Infrastructure vs. ad hoc modes

- Infrastructure mode基础设施模式：Hosts connect via base stations (which handle network services like addressing/routing).

- Ad hoc networks自组织网络 :No base stations; nodes only transmit to others within link coverage, and self-organize the network (route among themselves).

1.3.2 Handoff

When a mobile host moves between base station coverage areas, switching its connection point —this raises challenges (e.g., location tracking, seamless TCP/phone call continuity).

1.3.3 Wireless network classification (by hops + infrastructure)

This section introduces **network infrastructure** (the larger network wireless hosts communicate with) and a high-level classification framework for wireless networks, based on two criteria:

1. Whether packets traverse a single wireless hop or multiple hops.
 2. Whether the network includes infrastructure (e.g., base stations).
- Single-hop, infrastructure-based: Hosts connect to base stations (e.g., Wi-Fi, 4G LTE), which link to the larger Internet.
 - Single-hop, infrastructure-less: No base stations; no connection to the larger Internet (e.g., Bluetooth, ad hoc nets).
 - Multi-hop, infrastructure-based: Nodes relay via others to reach a base station (e.g., mesh nets, WSNs) and connect to the Internet.
 - Multi-hop, infrastructure-less: No base stations/Internet connection; nodes relay to reach others (e.g., MANET/VANET).Note: Ad hoc networks never became widely popular.

2. Wireless Link/Network Features

- **Wired → Wireless 替换**：以无线网卡（wireless NIC card）换有线以太网卡（wired Ethernet card）、接入点（access point, AP）换以太网交换机（Ethernet switch）；网络层（network layer）及以上无需改动，差异聚焦链路层（link layer）。
- **无线链路（wireless link）核心特征**：含信号衰减（signal attenuation，路径损耗 path loss）、干扰（interference，频段共享 / 环境噪声）、多路径传播（multipath propagation，信号失真）等问题；需权衡信噪比（signal-to-noise ratio, SNR）与误码率（bit error rate, BER），通过功率 / 物理层（调制方式）动态适配；还存在隐藏终端问题（hidden terminal problem）—— 节点互不可见时在共享节点处产生干扰。

2.1 CDMA 码分多址 Code Division Multiple Access

（需要一个好的复习链接）

3. WiFi: 802.11 无线局域网

IEEE 802.11 standard	Year	Max data rate	Range Indoors	Frequency	MU-MIMO
802.11a	1999	54 Mbps	30 m	5 GHz	No
802.11b	1999	11 Mbps	30 m	2.4 GHz	No
802.11g	2003	54 Mbps	30 m	2.4 GHz	No
802.11n (WiFi 4)	2009	600 Mbps	70 m	2.4 & 5 GHz	4x4 MIMO
802.11ac (WiFi 5)	2014	6.93 Gbps	70 m	5 GHz	8x8 MIMO Downlink
802.11ax (WiFi 6)	2020	9.6 Gbps	70 m	2.4 & 5 GHz	8x8 MIMO Uplink & Downlink
802.11af	2014	35 – 560 Mbps	1 Km	unused TV bands (54-790 MHz)	Yes
802.11ah	2017	347 Mbps	1 Km	900 MHz	Yes

- all use **CSMA/CA** for multiple access, and have **base-station** and **ad-hoc network versions**
- **MIMO (多输入多输出)**: Uses multiple antennas 天线 at **transmitter and receiver** (a smart antenna tech) to boost wireless throughput/range without extra bandwidth/power.
- **SISO/SIMO/MISO**: Degenerate MIMO cases:
 - MISO (多输入单输出): Receiver has 1 antenna.
 - SIMO (单输入多输出): Transmitter has 1 antenna.
 - SISO (单输入单输出): Neither has multiple antennas. (后期可删)

3.1 802.11 无线局域网体系结构

- **802.11 无线局域网架构**: 核心组件是**基本服务集 Basic Service Set (BSS, 基本服务集, aka “cell”)**, 包含无线站点与**接入点 (base station = access point (AP))**; 多个 BSS 通过交换机 / 路由器 (switches/routers) 接入互联网 (通常一个 AP + 路由器组成一个 BSS)。
- **MAC 地址**: 802.11 站点 / AP 使用 6 字节 MAC 地址 (由 IEEE 管理), 与以太网一致。
- **网络类型**:
 - **基础设施无线局域网 Infrastructure WLAN**: 依赖 AP (连接有线以太网)。
 - **自组织网络 Ad Hoc Network**: 无 AP, 站点直接互联 (适用于会议 / 车辆等临时无设施场景)。本节聚焦**基础设施无线局域网**。

3.1.1 信道与关联

- **802.11 关联**: 无线站点需与 AP 关联才能收发数据。
- **AP 配置**: 管理员为 AP 分配 **服务集标识符 (SSID)** 和信道 (2.4GHz 频段含 11 个重叠信道, 仅 1/6/11 无重叠, 同一网络最多部署 3 个 AP)。

- **WiFi 丛林**：存在多个 AP（不同子网 / 信道）的区域，站点需选择其中一个 AP 关联。
- **AP 扫描方式**：
 - 被动扫描：站点监听 AP 周期性发送的**信标帧**（含 SSID/MAC）。
 - 主动扫描：站点广播**探测请求帧**，从 AP 的探测响应中选择目标。

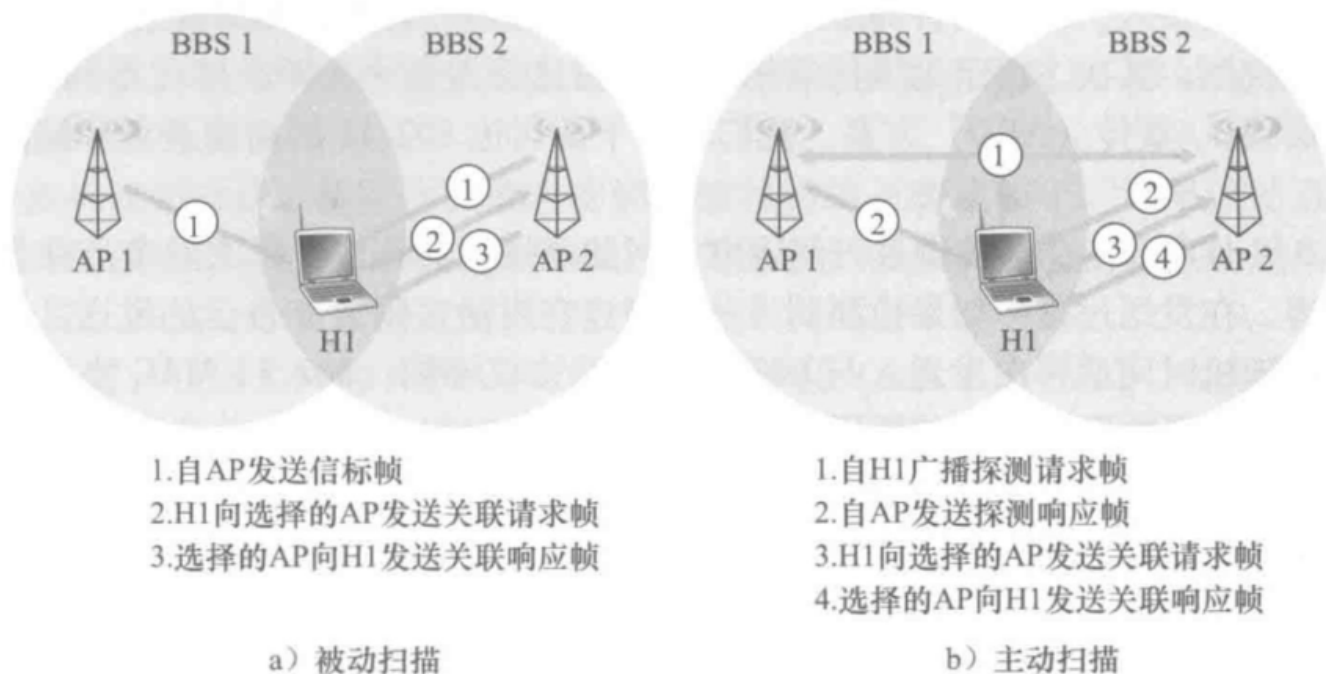


图 7-9 对接入点的主动和被动扫描

- **关联流程**：站点发送关联请求，AP 回复关联响应；随后站点通过该 AP 发送**DHCP**请求，获取对应子网的 IP 地址。
- **AP 鉴权**：采用 MAC 地址过滤或用户名 / 密码方式；AP 通过**RADIUS/Diameter**服务器实现集中式鉴权。

3.2 802.11 MAC 协议

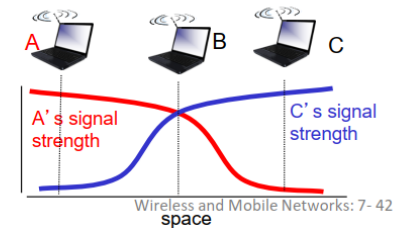
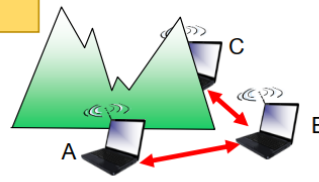
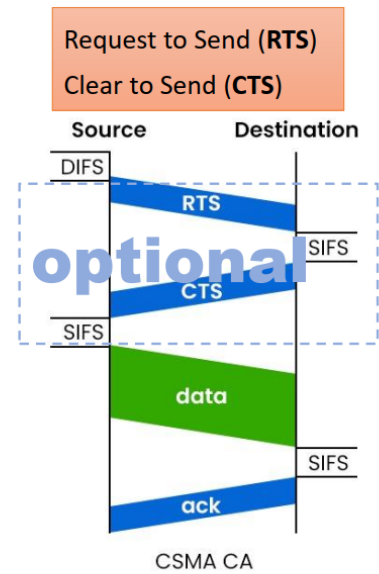
下面，我们将设备或者 AP 称为无线“站点”（station），它们共享多个接入信道。

宽泛地讲有三类多路访问协议：信道划分（包括 CDMA）、随机接入和轮流。

3.2.1 CSMA（CSMA with collision avoidance），或简称为 CSMA/CA

IEEE 802.11: multiple access

- **Goal: Collisions Avoidance** - let 2+ nodes NOT transmitting at same time (CSMA/CA)
- 802.11: CSMA - sense before transmitting
 - don't collide with detected ongoing transmission by another node
- 802.11: **no collision detection!**
 - **difficult to sense collisions:** high transmitting signal, weak received signal due to fading
 - The **default** and most common **method** is **CSMA/CA without RTS/CTS**.
 - **can't sense all collisions in any case:** hidden terminal, fading → can be solved via applying RTS/CTS



(我们需要非常棒的回顾链接)

3.3 IEEE 802.11 帧

3.4 在相同的 IP 子网中的移动性

4. 蜂窝网络：4G 和 5G



第一部分是Mobility